# Digital Twin for Smarter and Safer Work Zone: Proactive Intrusion Detection and Warning

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- USDOT / INDOT
- PSS Innovation
- Valtech Mobility
- Ramudden
- Indiana Sign & Barricade
- Butterfly Junction Technologies
- Aramsco

## 1. INTRODUCTION

- **Work zone intrusions** have been recognized as a safety-critical issue in the U.S.
- Work zone intrusion is a **multi-dimensional** issue, driven by a combination of **worker-related**, **vehicle-related**, and environmental factors

| *Across the **U.S.** (2022)* | | *Within the **Indiana** (2022)* | |
|---|---|---|---|
| 820 | work zone **fatal** crashes | 6348 | collisions within work zone |
| 890 | fatalities caused by crashes | 30.7 | fatalities **rate** / 1000 collisions |

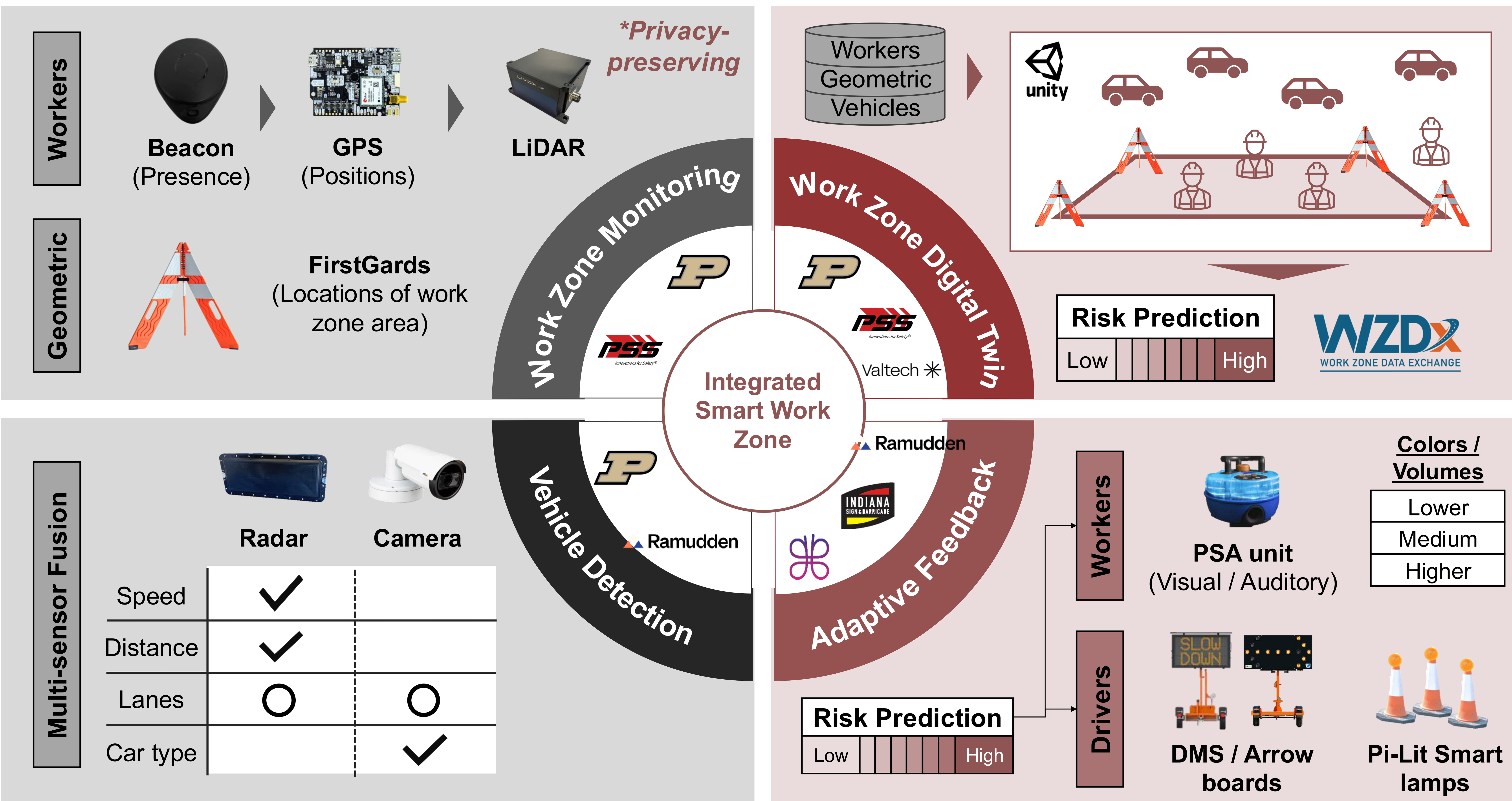
## 2. POINT OF DEPARTURE

- Despite existing mitigation technologies, **limitations** persist in addressing the **multi-dimensional** nature of the intrusions.
- An **integrated solution** is needed to thoroughly minimize work zone intrusions.

1. **No communication** among tech.
2. **Current** and **future** traffic conditions
3. **Static** and **mobile** work zones

Static + Mobile

## 3. PROPOSED INTEGRATED SOLUTION



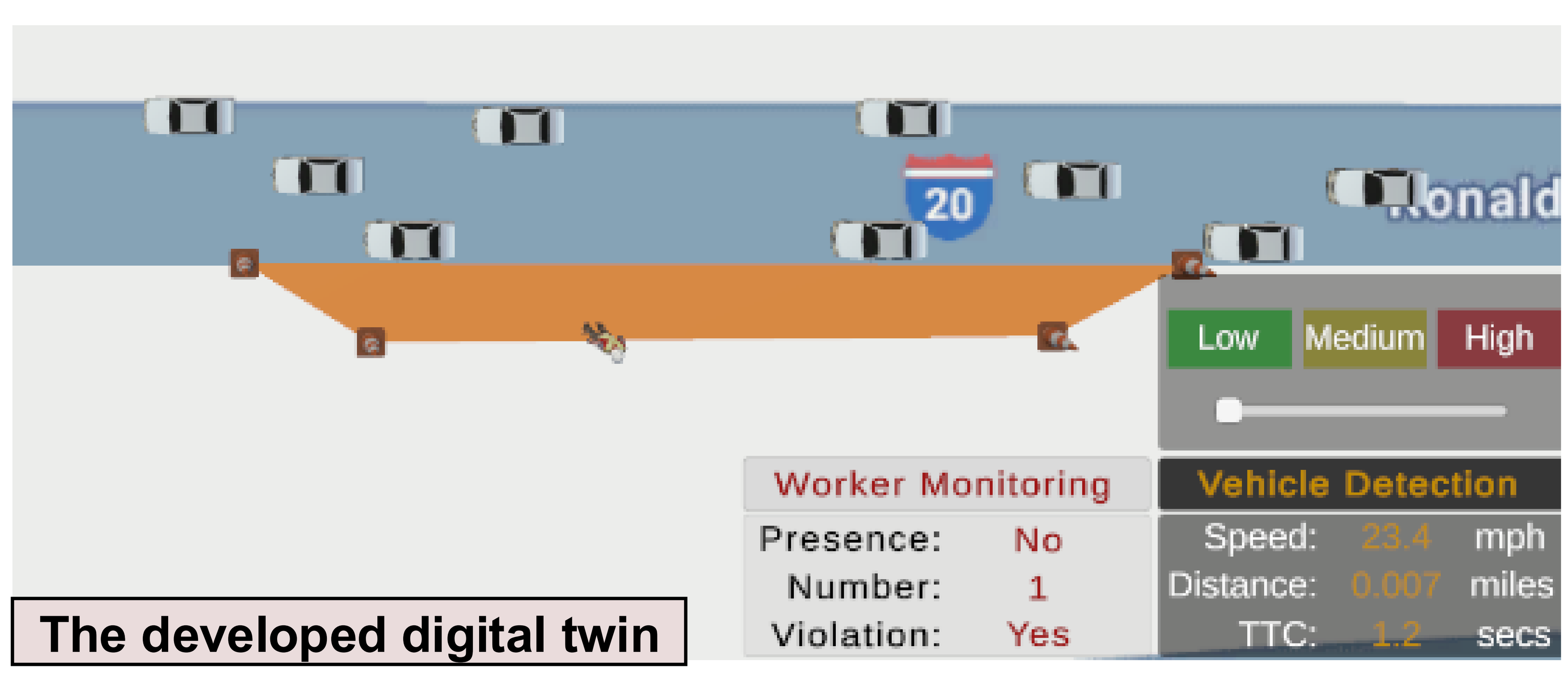

| | Radar | Camera |
|---|---|---|
| Speed | ✓ | |
| Distance | ✓ | |
| Lanes | ○ | ○ |
| Car type | | ✓ |

## 4. PROTOTYPE DEMONSTRATION IN ACTIVE WORKZONES

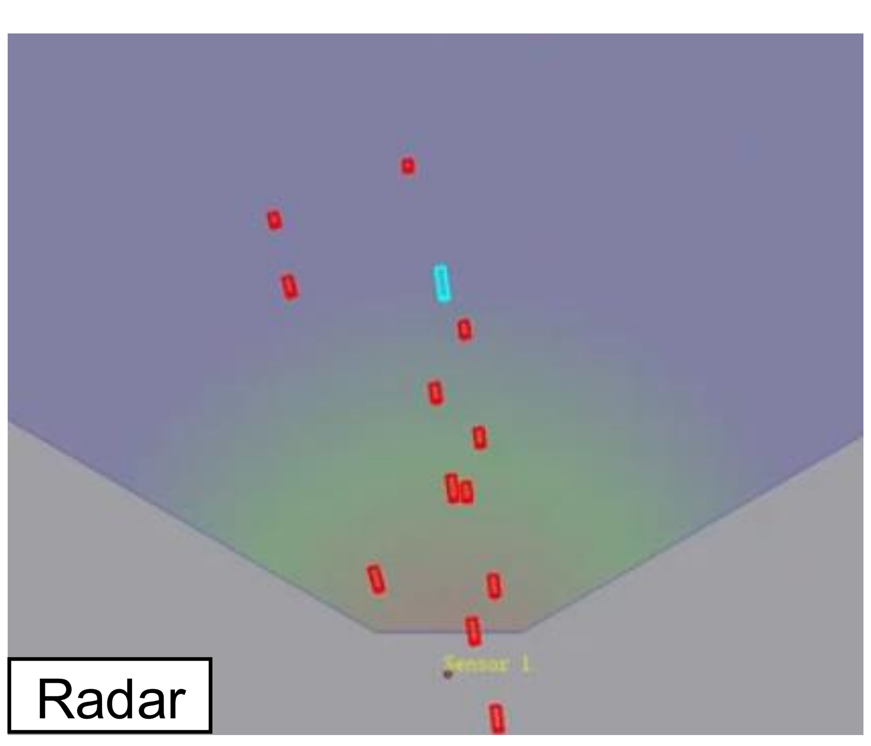

Sensor setup

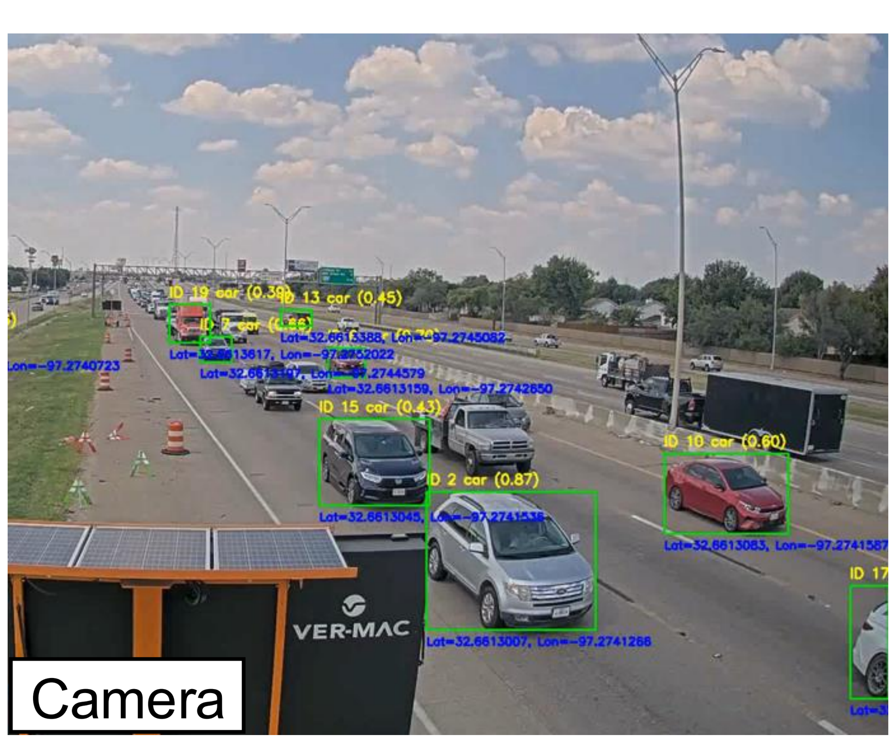

On-site deployment


FirstGards



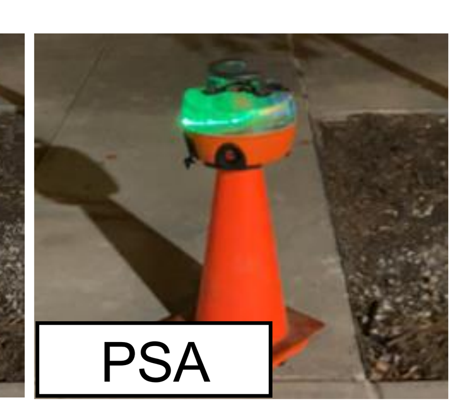
The developed digital twin


Radar


Camera


DMS


PSA


PSA


Pi-Lit

## 5. CONCLUSIONS

- The **pilot-test** showed that the system can accurately **generate geofence**, **monitor workers**, **detect vehicles**, and **offer feedback**.

## 6. ACKNOWLEDGMENT

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